APAC Country Risk Premiums (CRP)

Relative Volatility Multipliers and CRP for Asia-Pacific Markets

Jay Prakash

https://jay-finance.github.io/portfolio/

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Abstract The global CRP framework applies a single relative volatility multiplier of 3.03x uniformly across all 174 countries; a pragmatic simplification that works at the global level but breaks down when applied to markets as structurally different as Taiwan and Malaysia. Taiwan's sovereign bond market is institutionally dominated and structurally low-volatility, making its equity-bond ratio far higher than the global average implies. Malaysia's bond market absorbs a disproportionate share of macroeconomic uncertainty through foreign participation in local-currency debt, compressing its ratio well below 3.03x. Applying the same multiplier to both systematically understates Taiwan's CRP and overstates Malaysia's. This paper corrects that for ten APAC economies: China, Japan, India, South Korea, Australia, Indonesia, Singapore, Taiwan, Malaysia, and Thailand by replacing the global multiplier with country-specific relative volatility estimates. The resulting multipliers range from 1.74x in Malaysia to 8.14x in Taiwan. The direction of error is not uniform: Indonesia, Malaysia, Thailand, and Singapore all have country-specific multipliers below 3.03x, meaning the global model overstates their CRP. Australia converges almost exactly on the benchmark at 3.02x, serving as an intuitive sanity check on the methodology. The GDP-weighted APAC CRP is estimated at 3.25%, modestly above the global figure of 2.99%, driven primarily by China's dominant GDP weight and elevated multiplier.

1. Introduction

A single volatility multiplier applied globally is a pragmatic compromise. For most of the world, it is the only option: equity markets in sub-Saharan Africa or Central Asia lack the depth, duration, and data continuity needed to estimate a reliable equity-bond ratio. The global ACWI-to-bond ratio stands in for what would otherwise be a blank. It is better than nothing, but it is not especially precise.

The Asia-Pacific region does not face that constraint. Japan has one of the deepest equity markets in the world. Australia, South Korea, and Singapore sit firmly in developed-market territory by any classification. Even the emerging markets in this study: India, Indonesia, Malaysia, Thailand, maintain exchange-traded bond instruments with sufficient price histories to support meaningful volatility estimation. The data exists; using it is simply a matter of building the right inputs.

The global multiplier of 3.03x is derived from the ratio of ACWI equity volatility to JPMorgan Global Aggregate Bond Fund volatility over 2020–2026. It reflects the average relationship between equity and bond risk across 47 developed and emerging markets. For any individual country, that average is likely wrong in one direction or the other. Taiwan's bond market is exceptionally low-volatility, making its equity-bond ratio structurally higher than the global average. Malaysia's bond market is relatively high-volatility given the level of foreign participation in local-currency debt, compressing its ratio well below 3.03x. Applying the same multiplier to both countries introduces a systematic estimation error: Taiwan's CRP is understated, Malaysia's is overstated.

This paper corrects that. For each of the ten APAC markets in scope, the relative volatility multiplier is estimated using local instruments, the country's primary equity index paired against a liquid domestic sovereign bond index or ETF over the six-year window of March 2020 to March 2026.

Unless stated, all market data runs through 31 March 2026. CDS spreads are as of 1 April 2026.

2. Framework Recap

The full methodology is documented in the companion paper, "*Global Equity Risk Premiums (ERP) & Country Risk Premiums (CRP): Methodology Paper"* (Jay Prakash, May 2026). What follows is a summary sufficient to read this paper independently.

a. What is a Country Risk Premium?

Every valuation requires a discount rate, and every discount rate contains a risk premium, the extra return an investor demands above the risk-free rate to justify bearing equity risk. In a mature market like Denmark or Australia, that premium reflects the generic uncertainty of equity ownership and more precisely, it reflects what investors are collectively pricing into equities at this moment considering growth expectations, geopolitical noise, and

everything else embedded in current market prices. What it does not reflect are the additional risks that come with operating in a country where the political environment is unstable, violence is a real operating cost, corruption functions as an implicit tax, or the legal system cannot reliably enforce contracts.

The Country Risk Premium (CRP) is the incremental return investors require above the mature-market baseline to compensate for exactly those risks. Formally:

$$\textit{ERP (country)} = \textit{ERP (mature market)} + \textit{CRP (country)}$$

In the parent study we estimated, countries with zero sovereign risk arrive at exactly the mature-market ERP of 3.00%. Every basis point of CRP above that reflects the market's collective pricing of one or more of four risk dimensions: political structure, exposure to violence, corruption, and the quality of legal systems. None of these are niche academic constructs. Political risk is what happens when a newly elected government reverses the tax regime under which a business built its investment case. Exposure to violence drives up insurance costs, security costs, and the return premium that rational investors demand. Corruption is an implicit tax on revenue and a source of legal exposure that erodes value quietly over time. A legal system that cannot enforce contracts or property rights undermines the foundations of any business.

b. How is CRP Estimated?

The CRP is estimated in two steps. First, a country's sovereign default spread is extracted, Second, that default spread is multiplied by a relative volatility multiplier λ:

$$\textit{CRP} = \textit{Default Spread} \times \lambda$$

The multiplier exists because equity investors in the same country bear more risk than bondholders. A sovereign default wipes out bondholders, but equity investors absorb all residual losses first. The scale of that difference is what λ captures: for every 100 basis points of sovereign default spread, equity investors require λ × 100 basis points of additional return. In the parent study, a single global λ of 3.03x is applied uniformly. This paper replaces that uniform value with country-specific estimates for the APAC region.

c. Mature Market Baseline

This paper is concerned with CRP estimation and does not compute full equity risk premiums. That said, it is worth knowing how the CRP fits in: the mature-market ERP baseline is 3.00%, derived in the companion paper as the S&P 500 implied ERP of 4.00%

minus the United States' own CRP of roughly 1.00%. Any country's total ERP is simply that 3.00% plus whatever CRP this paper estimates for it. We won't be doing that addition here, but the building block is there if needed. One caveat worth flagging: the US CRP of 1.00% was itself estimated using the global relative volatility multiplier of 3.03x, not a US-specific relative volatility.

3. The Single Methodological Change: Country-Specific λ

In the parent study, the relative volatility multiplier is estimated using daily return data from the iShares MSCI ACWI ETF and the JPMorgan Funds Global Aggregate Bond Fund over March 2020 to March 2026, yielding a ratio of 3.03x applied uniformly across all 174 countries.

This paper substitutes that global ratio with country-specific ratios for the ten APAC markets covering 93.42% of total APAC's GDP. The construction logic is identical: annual standard deviations are computed from daily return series, the equity-to-bond ratio is calculated for each year, and the simple average (or trimmed average, where noted) over the full window is taken as λ for that country. The only change is the instruments used.

For each country, the equity proxy is the country's primary domestic index. The bond proxy is the most liquid available sovereign bond Index or ETF denominated in local currency. The period is March 31, 2020 to March 31, 2026.

Table 1: Instruments Used for Country-Specific λ Estimation

Country	Equity Index	Bond Instrument
China	CSI 300	CSOP China 5-Year Treasury Bond ETF (83199.HK)
Japan	Nikkei 225	iShares Core Japan Government Bond ETF (2561.T)
India	Nifty 50	Nifty 10-Year Benchmark G-Sec Index
South Korea	KOSPI 200	KOSEF 10-Year KTB ETF (148070.KS)
Australia	S&P/ASX 200	Vanguard Australian Fixed Interest Index ETF (VAF.AX)
Indonesia	IDX Composite	Premier ETF Indonesia Sovereign Bonds (XISB.JK)
Singapore	STI Index	ABF Singapore Bond Index Fund (A35.SI)
Taiwan	TAIEX	GreTai Taiwan Treasury Benchmark Index (TWTBI)
Malaysia	FTSE Bursa Malaysia KLCI	ABF Malaysia Bond Index Fund (0800EA.KL)
Thailand	SET Index	ABF Thailand Bond Index Fund (ABFTH.BK)

4. Annual Volatility Estimates and Country-Specific λ

Annual volatilities are computed year-by-year from daily returns following the same construction as the parent study. Daily log returns are calculated from adjusted closing prices, and annual standard deviations are derived by scaling daily standard deviations by the square root of the number of observed trading days. The annual relative volatility λ is the equity standard deviation divided by the bond standard deviation. The final λ is the simple average across all six annual observations, except where a trimmed mean is applied.

a. China

Table 2: Annual Relative Volatility - China

Period	Equity σ	Bond σ	λ
2020–21	20.59%	3.20%	6.44x
2021–22	17.66%	2.51%	7.02x
2022–23	17.87%	2.96%	6.04x
2023–24	13.91%	1.80%	7.71x
2024–25	20.67%	2.42%	8.54x
2025–26	15.10%	2.30%	6.56x
Average	**17.63%**	**2.53%**	**6.96x**

China's relative volatility estimates remain broadly stable across the sample, with λ largely clustering between 6x and 8x. The only clear outlier is 2024–25, where relative volatility rises to 8.54x. The increase appears primarily equity-driven rather than the result of abnormal bond-market behaviour. CSI 300 volatility rose sharply from 13.91% to 20.67%, while bond volatility remained within its historical range of roughly 2–3%. This suggests that the widening in λ reflects elevated uncertainty in Chinese equities rather than distortion in the sovereign bond proxy.

The period coincided with continued weakness in China's property sector, including ongoing stress among major developers such as Evergrande, Country Garden, and Vanke, alongside weak domestic demand and concerns surrounding long-term growth expectations. At the same time, Chinese government bond markets remained relatively stable and policy-anchored, with yields trading in unusually narrow ranges. The result was a temporary widening between equity-market risk pricing and sovereign bond stability, pushing the country-specific λ materially above the global benchmark of 3.03x. Despite

the elevated 2024–25 observation, the broader six-year pattern remains internally consistent, supporting the robustness of the estimated average λ of 6.96x.

b. Japan

Table 3: Annual Relative Volatility - Japan

Period	Equity σ	Bond σ	λ
2020–21	20.01%	2.47%	8.11x
2021–22	19.69%	2.19%	8.99x
2022–23	17.60%	4.97%	3.54x
2023–24	16.31%	6.40%	2.55x
2024–25	25.89%	5.06%	5.12x
2025–26	27.50%	6.97%	3.94x
Average	**21.17%**	**4.68%**	**4.53x**

Japan's elevated relative volatility during 2020–22 appears to have been driven primarily by unusually suppressed sovereign bond volatility rather than exceptionally high equity volatility. Equity volatility during the period remained relatively stable near 20%, broadly consistent with other developed markets during the COVID-19 shock. The more important factor was the extremely low volatility of Japanese government bonds under the Bank of Japan's Yield Curve Control (YCC) framework, which kept the 10-year JGB yield pinned near 0% through large-scale bond purchases and fixed-rate purchase operations. In 2021–22, for example, bond volatility was compressed to just 2.19% while equity volatility remained at 19.69%, producing a ratio of 8.99x.

From late 2022 onward, relative volatility normalised as bond volatility repriced materially higher. The Bank of Japan widened the allowable trading band for 10-year JGB yields December 2022, increased policy flexibility further through 2023, and ultimately exited YCC in March 2024. As a result, bond volatility rose from roughly 2% during 2020–22 to a 5–7% range thereafter. Equity volatility also increased during this period, reflecting global macro uncertainty, yen weakness, and inflation repricing, but the increase in bond volatility was proportionally larger, compressing the equity-to-bond ratio into a more stable post-2022 range. The unusually high λ observations during 2020–22 should therefore be interpreted primarily as a consequence of artificially suppressed JGB volatility under BOJ policy rather than a structurally abnormal increase in Japanese equity risk.

c. India

Table 4: Annual Relative Volatility - India

Period	Equity σ	Bond σ	λ
2020–21	22.43%	4.52%	4.97x
2021–22	15.70%	3.33%	4.72x
2022–23	14.66%	4.99%	2.94x
2023–24	9.69%	2.65%	3.66x
2024–25	13.97%	2.15%	6.51x
2025–26	13.51%	3.33%	4.06x
Average	**14.99%**	**3.49%**	**4.29x**

India's volatility structure is one of the cleaner datasets in the sample, but the six-year period contains three distinct phases that the summary average obscures. The first phase covers 2020–22 where equity volatility remained elevated in 2020-21 at 22.43%, reflecting the pandemic-driven disruption and subsequent recovery. Bond volatility was comparatively contained at 4.52%, producing a relative volatility of 4.97x. Both measures normalised in 2021–22 as markets stabilised, with λ holding broadly steady near 4.72x.

The second phase covers 2022–23 and marks the RBI tightening cycle. The Reserve Bank of India began hiking the repo rate aggressively from May 2022, starting with an unscheduled 40 basis point increase and raised it from 4.00% to 6.50% by February 2023. The sharp repricing in Indian government bond yields, which moved from approximately 6.84% at end-March 2022 to around 7.45% by June 2022, pushed bond volatility up to 4.99%. At the same time, equity volatility continued declining to 14.66% as Indian markets absorbed the rate shock relatively well given robust earnings growth and domestic demand. The combination compressed lambda to 2.94x, its lowest point in the sample.

The third phase covers 2023–25 and is dominated by two overlapping forces. In 2023–24, the RBI held rates steady at 6.50% throughout the year, removing a major source of bond market uncertainty. Simultaneously, JPMorgan's September 2023 announcement of India's inclusion in the GBI-EM index beginning June 28, 2024 triggered pre-positioning inflows, foreign investors had already channelled roughly $10 billion into eligible securities ahead of the formal inclusion date, with yields declining from approximately 7.20% at the start of 2024 to below 7.00% by mid-year. The combined effect of rate stability and index-driven demand compressed both equity volatility at 9.69% and bond volatility at 2.65% to their joint lows across the sample, producing a λ of 3.66x.

In 2024–25, the GBI-EM inclusion formally commenced and ran over ten months, drawing an estimated $22–30 billion in passive and active inflows into Indian government securities over the inclusion window. Increased demand from both domestic institutions and global index-tracking funds contributed to a further compression in bond volatility to just 2.15%, the lowest single-year bond volatility observation for India in this study, while equity volatility remained at 13.97%. The result was a sharp λ spike to 6.51x, mechanically driven by the bond side rather than any change in equity-market conditions.

The compression appears temporary. Bond volatility normalised back to 3.33% in 2025–26 as the inclusion window closed and the marginal flow effect faded, with λ settling to 4.06x. The overall lambda estimate of 4.29x is therefore best understood as the average of three distinct regimes: a tightening-cycle compression, a dual-calm pre-inclusion period, and a structural baseline that the 2025–26 observation likely best represents going forward.

d. South Korea

Table 5: Annual Relative Volatility - South Korea

Period	Equity σ	Bond σ	λ
2020–21	22.56%	3.55%	6.36x
2021–22	15.42%	5.22%	2.95x
2022–23	18.60%	9.54%	1.95x
2023–24	15.65%	6.94%	2.25x
2024–25	20.94%	4.90%	4.27x
2025–26	34.74%	4.92%	7.07x
Average	**21.32%**	**5.84%**	**3.65x**

South Korea exhibits two distinct volatility regimes over the sample period. During 2021–24, relative volatility compressed sharply as sovereign bond volatility increased materially while equity volatility declined. The primary driver was the sharp rise in Korean government bond volatility, from 3.55% in 2020–21 to 9.54% in 2022–23. This coincided with the global inflation shock and aggressive monetary tightening cycle following the pandemic: the Bank of Korea raised policy rates rapidly between 2021 and 2023 as inflation accelerated and the Korean won weakened against the US dollar. Korean bond markets were particularly sensitive to global rates given Korea's open capital markets and export-oriented economy, resulting in unusually elevated sovereign bond volatility during the period. At the same time, equity volatility moderated from COVID-era highs, contributing further to the compression in λ.

The pattern reversed sharply in 2025–26. Bond volatility normalised back toward a ~5% range, while equity volatility increased to 34.74%, the highest single-year observation in this study for any country. South Korean equities became significantly more volatile amid renewed instability in the semiconductor cycle, heightened sensitivity to global AI-related technology concentration, and the domestic political crisis that erupted in December 2024 when President Yoon Suk-yeol declared emergency martial law. The resulting constitutional crisis: Yoon's impeachment, arrest, and subsequent conviction for insurrection, sustained elevated political uncertainty through the snap presidential election

held in June 2025. Because bond volatility simultaneously declined from its 2022–23 peak, relative volatility rebounded sharply, producing the 7.07x observation in 2025–26.

e. Australia

Table 6: Annual Relative Volatility - Australia

Period	Equity σ	Bond σ	λ
2020–21	19.40%	3.87%	5.01x
2021–22	12.46%	4.00%	3.11x
2022–23	15.19%	6.75%	2.25x
2023–24	10.51%	4.78%	2.20x
2024–25	11.81%	4.31%	2.74x
2025–26	13.18%	3.58%	3.68x
Average	**13.76%**	**4.55%**	**3.02x**

Australia's volatility structure is one of the most stable and economically coherent in the sample. The elevated 2020–21 observation primarily reflects the COVID shock: ASX 200 volatility remained elevated during the pandemic recovery phase, particularly given Australia's exposure to global commodity demand and cyclical sectors, while bond volatility remained comparatively contained under accommodative monetary policy and RBA intervention. From 2021 onward, relative volatility compressed steadily as sovereign bond volatility increased materially during the global inflation and rate-hike cycle. Australian government bond yields repriced sharply higher following aggressive RBA tightening beginning in 2022, producing elevated fixed-income volatility through 2022–24.

The result was a relatively stable post-2022 range of roughly 2x–3.5x, with both equity and bond volatility moving broadly in tandem. Unlike Japan, there was no artificial suppression of bond volatility, and unlike emerging markets such as Indonesia or Thailand, Australia avoided major currency or fiscal stress. At 3.02x, Australia's country-specific λ is virtually identical to the global benchmark of 3.03x. The close convergence appears economically intuitive and reinforces Australia's role as a stable developed-market reference point within the APAC sample.

f. Indonesia

Table 7: Annual Relative Volatility - Indonesia

Period	Equity σ (%)	Bond σ (%)	λ
2020–21	19.57%	5.38%	3.63x
2021–22	11.75%	2.81%	4.18x
2022–23	12.95%	5.09%	2.55x
2023–24	8.70%	4.62%	1.88x
2024–25	16.75%	7.66%	2.19x
2025–26	20.20%	129.06%	0.16x
Trimmed Avg.	**-**	**-**	**2.56x**

2025–26 observation highlighted: bond volatility of 129.06% is treated as an outlier and excluded from the average. λ is estimated using a trimmed mean of 34% that removes the minimum (0.16x) and the maximum (4.18x) of observations across the six-year sample, yielding a trimmed average of 2.56x.

Indonesia's relative volatility estimates are broadly stable between 2020–25, generally fluctuating within a 2x–4x range. The 2025–26 observation, however, is clearly anomalous: bond volatility surged from 7.66% to 129.06% while equity volatility increased only modestly from 16.75% to 20.20%, collapsing the implied ratio to 0.16x. A move of this magnitude is inconsistent with normal sovereign bond market behaviour.

The period coincided with significant pressure on Indonesian financial markets. The rupiah weakened to multi-decade lows against the US dollar amid concerns surrounding fiscal credibility, widening deficits under the Prabowo administration, and uncertainty surrounding costly social programmes including the flagship free meals scheme. Foreign investors reduced exposure to Indonesian government bonds as market concerns intensified: overseas investors pulled an estimated $6.4 billion from Indonesia's sovereign bond market in 2025 alone, with the largest outflows following President Prabowo's abrupt removal of Finance Minister Sri Mulyani Indrawati. The 2025 budget deficit reached 2.92% of GDP, the widest in at least two decades outside the pandemic. While these stresses were real, the magnitude of the bond volatility spike in the ETF proxy makes the raw 2025–26 observation unsuitable as a structural estimate, and the trimmed average of 2.56x is used as the final λ for Indonesia.

g. Singapore

Table 8: Annual Relative Volatility - Singapore

Period	Equity σ	Bond σ	λ
2020–21	16.32%	4.03%	4.05x
2021–22	12.03%	4.02%	2.99x
2022–23	10.81%	5.90%	1.83x
2023–24	9.55%	4.72%	2.02x
2024–25	10.06%	5.58%	1.80x
2025–26	14.30%	4.62%	3.09x
Average	**12.18%**	**4.81%**	**2.53x**

Singapore's volatility structure is among the most stable in the sample and closely resembles that of a mature developed market. Relative volatility remained within a relatively narrow 2x–4x range throughout the period, with no major structural distortions or abnormal observations. The elevated 2020–21 observation primarily reflected the COVID-era equity shock: STI volatility remained elevated during the pandemic recovery period while bond volatility stayed relatively contained under Singapore's highly managed monetary and financial framework.

From 2021 onward, relative volatility compressed gradually as equity volatility declined and bond volatility increased moderately during the global inflation and tightening cycle. The Monetary Authority of Singapore's exchange-rate-based monetary framework and strong sovereign credit profile helped stabilise domestic financial conditions throughout the period. The STI's sector composition, heavily weighted toward financials, industrials, and real estate rather than high-beta technology sectors, helping keep equity volatility within a relatively narrow 9–16% range. The result is a structurally low average λ of 2.53x, reflecting Singapore's relatively balanced equity-bond volatility relationship and producing one of the most internally consistent datasets in the study.

h. Taiwan

Table 9: Annual Relative Volatility - Taiwan

Period	Equity σ	Bond σ	Λ
2020–21	16.00%	0.92%	17.40x
2021–22	16.54%	4.85%	3.41x
2022–23	19.26%	2.38%	8.08x
2023–24	11.76%	1.02%	11.56x
2024–25	21.74%	2.71%	8.03x
2025–26	25.21%	1.70%	14.86x
Average	**18.42%**	**2.26%**	**8.14x**

Taiwan's volatility structure is dominated by persistently low sovereign bond volatility combined with increasingly volatile equity markets. Unlike Japan or Indonesia, the unusually high λ observations are not driven by temporary dislocations, but by a structural divergence between Taiwan's highly stable government bond market and its technology-heavy equity market. The 2020–21 observation of 17.40x was primarily the result of exceptionally low bond volatility at just 0.92%, coinciding with aggressive global monetary easing and strong demand for high-quality sovereign fixed income during COVID. Taiwan's bond market is institutionally dominated, domestically held, and subject to active central bank stabilisation, all of which contribute to structurally low volatility.

The sharp decline in λ during 2021–22 to 3.41x was driven almost entirely by a repricing in bond volatility from 0.92% to 4.85% during the global inflation shock, while equity volatility remained broadly unchanged at 16.54%. From 2022 onward, bond volatility normalised lower again toward a 1–3% range while equity volatility increased materially. The TAIEX became increasingly sensitive to the global semiconductor and AI cycle, given its heavy concentration in Taiwan Semiconductor Manufacturing Company (TSMC) and large-cap technology exporters. Rising geopolitical tensions surrounding Taiwan and continued volatility in global technology demand further amplified equity-market swings during 2024–26, producing the 14.86x observation in 2025–26. The persistence of extremely low bond volatility throughout most of the sample suggests that Taiwan's elevated average λ of 8.14x is structural rather than anomalous.

i. Malaysia

Table 10: Annual Relative Volatility - Malaysia

Period	Equity σ	Bond σ	λ
2020–21	15.12%	6.89%	2.19x
2021–22	10.53%	5.51%	1.91x
2022–23	11.59%	7.33%	1.58x
2023–24	6.90%	3.62%	1.91x
2024–25	10.41%	8.96%	1.16x
2025–26	11.77%	5.87%	2.00x
Average	**11.05%**	**6.36%**	**1.74x**

Malaysia's relative volatility remains consistently compressed within a narrow 1.0x–2.2x range, reflecting the fact that Malaysian sovereign bond volatility remained persistently high relative to equity volatility throughout the period. The most notable observation occurs in 2024–25, where λ falls to its lowest level of 1.16x. Unlike Taiwan or China, the compression was not driven by unusually calm equities but by a sharp increase in bond-market volatility to 8.96%, substantially above historical norms for most Asian sovereign debt markets. This coincided with persistent global rate uncertainty, fluctuations in US Treasury yields, and sustained pressure on emerging-market currencies and bond markets following the post-pandemic tightening cycle. Malaysia's bond market remained particularly sensitive to foreign capital flows given the relatively high level of foreign participation in local-currency government debt.

At the same time, KLCI equity volatility remained comparatively subdued throughout the sample. The FTSE Bursa Malaysia KLCI is heavily weighted toward financials, plantations, and commodity-linked firms, sectors that generally exhibit lower realised volatility than technology-heavy benchmarks. The persistence of relatively high bond volatility across the sample explains Malaysia's structurally low average λ of 1.74x.

The result reflects that a larger share of Malaysia's macroeconomic and policy uncertainty is absorbed through the sovereign bond market than in most other APAC economies. Hence, mechanically compressing the equity-to-bond ratio relative to markets where bond volatility is structurally contained.

j. Thailand

Table 11: Annual Relative Volatility - Thailand

Period	Equity σ	Bond σ	λ
2020–21	19.01%	5.99%	3.17x
2021–22	11.30%	6.25%	1.81x
2022–23	10.83%	9.32%	1.16x
2023–24	11.04%	3.44%	3.21x
2024–25	13.16%	8.22%	1.60x
2025–26	19.29%	8.45%	2.28x
Average	**14.11%**	**6.95%**	**2.03x**

Thailand's volatility structure is relatively stable overall, but unlike Malaysia, the fluctuations in λ are driven by alternating shifts in both equity and bond volatility rather than one persistent structural factor. The sharpest compression occurs during 2022–23, where bond volatility rose above 9% during the global inflation and tightening cycle as emerging-market fixed-income markets repriced sharply higher yields following aggressive US Federal Reserve rate hikes. Thailand's bond market was additionally affected by capital outflows, baht weakness, and uncertainty surrounding the pace of domestic economic recovery, particularly given Thailand's heavy dependence on tourism and external demand. At the same time, Thai equity volatility remained comparatively subdued near 11%, reflecting slower domestic growth but limited speculative excess relative to technology-heavy Asian equity markets.

The ratio rebounded in 2023–24 as bond volatility normalised lower, coinciding with improving tourism flows and greater stability in regional bond markets as inflation pressures eased globally. However, bond volatility rose again during 2024–26 amid renewed fiscal uncertainty, political instability, and continued sensitivity to global rates markets. Thailand experienced repeated changes in coalition dynamics and policy uncertainty during this period, contributing to persistent pressure on domestic financial markets. Thailand's relatively low average λ of 2.03x appears to reflect structurally elevated sovereign bond volatility rather than unusually low equity risk: much of the country's macroeconomic and policy uncertainty was transmitted through the bond market during the sample period, compressing the equity-to-bond ratio relative to the broader APAC average.

5. Summary: Country-Specific λ vs. Global Benchmark

The table below consolidates the country-specific λ estimates alongside the global benchmark of 3.03x, ordered from highest to lowest multiplier. The range is striking: from 1.74x in Malaysia to 8.14x in Taiwan. Four countries: China, Taiwan, Japan, and India, sit meaningfully above the global benchmark, meaning the parent study systematically understates their CRP. Three countries: Indonesia, Malaysia, Thailand, and Singapore sit below it, where the parent study overstates. Australia and Singapore effectively converge on the global estimate.

Table 12: Country-Specific λ vs. Global Benchmark (3.03x)

Country	Country-Specific λ	Difference
Taiwan	8.14x	+5.11x
China	6.96x	+3.93x
Japan	4.53x	+1.50x
India	4.29x	+1.26x
South Korea	3.65x	+0.62x
Australia	3.02x	-0.01x
Indonesia[1]	2.56x	-0.47x
Singapore	2.53x	-0.50x
Thailand	2.03x	-1.00x
Malaysia	1.74x	-1.29x

6. Estimating Default Spread

Default spreads are estimated using one of two methods depending on data availability. For most countries in this study, 10-year CDS markets exist and are liquid enough to serve as the primary input. Where CDS data is unavailable, a rating-based fallback is applied.

Method A: CDS-Based Default Spread

Where 10-year sovereign CDS markets exist, they are used directly. CDS spreads are market-determined and forward-looking, repricing in real time as sovereign risk shifts. Switzerland's CDS spread is netted out to isolate the country-specific component:

$$\textit{Default Spread (CDS)} = \textit{CDS country} - \textit{CDS Switzerland}$$

Switzerland’s CDS of 0.17% is chosen as the risk-free sovereign benchmark as we assume it as the world’s safest sovereign with trading CDS spread, reflecting near-zero default probability, political neutrality, and deep capital markets.

Table 13: Credit Default Swaps

Country	10Y CDS Spread	Net of Switzerland
China	0.77%	0.60%
Japan	0.51%	0.34%
India	1.02%	0.85%
South Korea	0.50%	0.33%
Australia	0.25%	0.08%
Indonesia	1.59%	1.42%
Malaysia	0.78%	0.61%
Thailand	0.89%	0.72%

Method B: Rating-Based Default Spread

In this study, Singapore and Taiwan does not have a10-year CDS data, a Moody's rating-based lookup is applied instead. Rather than using Damodaran's published rating-to-spread table, the spreads here are derived from the average CDS of all countries within each Moody's rating category. This produces a market-implied spread for each letter grade that is internally consistent with the CDS data used in Method A, rather than drawing from an external source with a different construction methodology.

Table 14: Rating-Based Average Spreads

Country	Moody's Rating	Average CDS Spread	Net of Switzerland
Singapore	Aaa	0.22%	0.05%
Taiwan	Aa3	0.58%	0.41%

7. Country Risk Premium Estimates

With country-specific λ values in hand, the CRP for each country is computed following the standard framework. The default spread is taken from CDS and average when CDS not available for all ten countries, with Switzerland's CDS spread of 0.17% netted out to isolate the country-specific component.

Table 15: APAC CRP Estimates

Country	Default Spread	λ	CRP[1]	CRP[2]
China	0.60%	6.96x	4.18%	1.82%
Japan	0.34%	4.53x	1.54%	1.03%
India	0.85%	4.29x	3.65%	2.58%
South Korea	0.33%	3.65x	1.20%	1.00%
Australia	0.08%	3.02x	0.24%	0.24%
Indonesia	1.42%	2.56x*	3.64%	4.30%
Singapore	0.05%	2.53x	0.13%	0.15%
Taiwan	0.41%	8.14x	3.33%	1.24%
Malaysia	0.61%	1.74x	1.06%	1.85%
Thailand	0.72%	2.03x	1.46%	2.18%

1. Country Risk Premium Estimated using Country Specific λ

2. Country Risk Premium Estimated using Global λ Benchmark of 3.03x

The final column, CRP^2: shows what the parent study's uniform 3.03x multiplier would have produced for each country using the same CDS inputs. The divergences are most pronounced at the extremes: Taiwan's CRP is 3.33% under the country-specific model versus 1.24% under the global model, a gap of 209 basis points. China's CRP is 4.18% versus 1.82%, a difference of 236 basis points. These are not rounding errors; they are the difference between a cost of equity that adequately compensates for the equity risk of operating in these markets and one that does not. At the other extreme, Malaysia and Thailand carry lower CRPs under the country-specific model than the global model, the direction of error reverses because bond volatility absorbs a larger fraction of macro risk in those markets than the global average implies.

8. APAC Regional Aggregate

Country-level estimates are aggregated to a regional average using GDP weighting, consistent with the methodology in the parent study. China's weight dominates, which means China's elevated λ and CRP pull the regional estimate meaningfully above what an equal-weighted average would produce.

Table 16: GDP-Weighted APAC Regional CRP

Country	GDP (USD mn)	CRP
China	18,743,803	4.18%
Japan	4,027,598	1.54%
India	3,909,892	3.65%
South Korea	1,875,388	1.20%
Australia	1,757,022	0.24%
Indonesia	1,396,300	3.64%
Singapore	547,387	0.13%
Taiwan	801,495	3.33%
Malaysia	422,227	1.06%
Thailand	526,518	1.46%
APAC	**34,007,629**	**3.25%**

The GDP-weighted APAC CRP of 3.25% compares to the parent study's global CRP of 2.99%. The APAC region carries modestly higher aggregate country risk than the world average, driven primarily by China's large GDP weight and elevated λ, a directionally correct premium for a region with concentrated geopolitical and structural risk in its largest economy.

For reference, the parent study's global model using a uniform 3.03x across these same ten markets with the same CDS inputs would produce a GDP-weighted APAC CRP of approximately 1.75%. The country-specific model adds approximately 150 basis points to that estimate, almost entirely attributable to China's elevated multipliers.

9. Conclusion

The case for country-specific relative volatility multipliers in APAC is straightforward: the data exists, the markets are liquid enough to support it, and the difference from the global benchmark is too large to ignore for at least four of the ten countries studied. Taiwan and China in particular carry multipliers that are nearly double the global average, meaning the parent study's CRP estimates for those markets are materially understated from an equity investor's perspective.

The results are not uniformly in one direction. Indonesia, Malaysia, Thailand, and Singapore all have country-specific λ values below the global benchmark, meaning the parent study overstates their CRP. This is consistent with the structure of those bond

markets, which are liquid enough to absorb a meaningful share of macro risk and mechanically compresses the equity-to-bond ratio. Two markets: Australia and South Korea, sit close enough to the global benchmark that the choice of multiplier is largely immaterial in practice.

The CRP estimates here are relevant where country risk is a genuine driver of value. A company whose operations, cash flows, and regulatory exposure are concentrated in a single APAC market should carry that country's CRP in full. For multinationals or companies with geographically diversified revenue, the relevant risk premium is a blend weighted by where value is actually generated, not simply the CRP of the country of incorporation. The estimates here provide the building blocks for that calculation; the judgment of how much country risk a specific a company, project or an investment actually bears remains with the analyst.

10. Key Assumptions

a. Country-specific λ replaces global λ for APAC only

The parent study's uniform multiplier of 3.03x continues to apply to all countries outside this study's scope. No other input in the framework is modified.

b. Local instrument selection

The equity index and bond proxy for each country are chosen to best represent domestic investable risk. For countries where multiple bond ETF options exist, the largest and most liquid by AUM and trading volume is selected. The choice of instrument can affect the measured λ.

c. Trimmed mean for Indonesia

A 34% trimmed mean is applied for Indonesia to remove the anomalous 2025–26 bond volatility observation. The trimming removes the minimum and maximum values from the six-year λ series, yielding a trimmed average of 2.56x.

d. CDS spreads remain the default spread input

Eight countries in scope have 10-year CDS markets (Method A). Singapore and Taiwan do not, so Method B (rating-based averages) is used.

e. GDP weighting for regional aggregation

GDP weights reflect the 2024 World Bank estimates used in the parent study.

11. Limitations

a. Instrument liquidity

Bond ETF liquidity varies materially across this sample. For India Nifty G-Sec index is used and TWTBI for Taiwan, these are price indices and not a total return instrument, introducing a minor comparability issue relative to dividend-adjusted equity returns.

b. Structural breaks

The study window encompasses the Bank of Japan's yield curve control period and its subsequent unwinding, which created artificial suppression and then release of JGB volatility. Japan's λ of 4.53x should be understood in that context. In a fully normalised rate environment, the structural ratio may be somewhat lower.

c. Taiwan geopolitical premium

Taiwan's high λ reflects the structural low-volatility of its bond market as much as geopolitical risk. An analyst who believes the geopolitical risk premium for Taiwan is incompletely captured by CDS markets may wish to layer additional adjustments on top of the country-specific CRP.

d. CDS vintage

CDS spreads are sourced from Damodaran Online as of 1st of April 2026. Users are encouraged to update CDS inputs for any country where conditions may have changed since that date.

12. Data Sources

Table 17: Data Sources

Data Item	Source	Period
APAC Equity Index Returns	Yahoo Finance	Mar 2020 – Mar 2026
APAC Bond ETF / Index Returns	Yahoo Finance	Mar 2020 – Mar 2026
10-Year CDS Spreads	NYU Stern / Damodaran Online	April 1st, 2026
Mature Market ERP (3.00%)	S&P 500 Implied ERP - parent methodology paper	March 2026
Country GDP	World Bank World Development Indicators	2024

13. References

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Bank of Japan (2022): Modification of Yield Curve Control, December 20, 2022.

JPMorgan Asset Management (2024): India's Inclusion in the GBI-EM Index, June 2024.

Reuters (2024–2026), Indonesia: Rupiah, fiscal policy, and Bank Indonesia intervention.

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